ViBridge-VQA: Bridging Global and Local Visual Tokens for Parameter-Efficient Vietnamese Visual Question Answering

Nguyen Trung Quoc, Nguyen Anh Tuong, Phan Ba Duc, Tran Dac Thinh, Dang Duy Lan, Nguyen Quoc Thinh, and Tung Le (✉)

Faculty of Information Technology, University of Science, Ho Chi Minh City, Vietnam
Vietnam National University, Ho Chi Minh City, Vietnam
{22120071,22120182,22120301,22120347,
22120348,22120412}@student.hcmus.edu.vn, lttung@fit.hcmus.edu.vn

Abstract. Vision–language models (VLMs) such as Vintern-1B have brought strong multimodal abilities to Vietnamese, but adapting them to a new Visual Question Answering (VQA) benchmark remains costly: the usual recipe fine-tunes the language model on many high-resolution image tiles, while task-specific architectures must be trained from scratch. In this paper, we propose **ViBridge-VQA**, a parameter-efficient approach that improves Vintern-1B without updating any of its pre-trained weights. ViBridge-VQA feeds the frozen language decoder two complementary kinds of visual tokens from a single image view: *global* tokens, produced by a small trainable bridge from the `[CLS]` vector that Vintern-1B discards, and *local* tokens, produced by Vintern-1B's own pre-aligned projector and pooled to a chosen budget. We treat the number of tokens as an accuracy–cost trade-off and select it on validation data as the knee of the Pareto front. On the reasoning-oriented AutoViVQA benchmark, the selected budget of 50 visual tokens gives two main results. Training only the bridge (1.35% of the parameters) raises the F1 of Vintern-1B from 17.6 to 54.6, comparable to the published F1 of the fine-tuned Vintern-1B (53.8), whose decoder reads 1,792 visual tokens per image, 36 times more than ours. Training a lightweight decoder LoRA together with the bridge (1.57% of the parameters) raises F1 further to 58.2. At a fixed number of global tokens, adding local tokens improves F1 far more than increasing the global tokens from 8 to 14, and on four other Vietnamese VQA datasets the model with local tokens outperforms a global-only model with a decoder LoRA.

Keywords: Vietnamese Visual Question Answering · Vision–Language Models · Parameter-Efficient Learning · Visual Token Budget · Generative VQA

(✉)Corresponding author: Tung Le – lttung@fit.hcmus.edu.vn

1 Introduction

Visual Question Answering (VQA) asks a model to answer a natural-language question about an image. The task has progressed rapidly with large vision–language models (VLMs) such as BLIP-2 [12], Flamingo [1] and LLaVA [19], which connect a pre-trained vision encoder to a pre-trained language model and generate free-form answers. Vietnamese has followed this trend: Vintern-1B [7] connects an InternViT-300M encoder [6] to a Qwen2.5-0.5B decoder [27] and is an open Vietnamese VLM, while benchmarks such as ViVQA [30], OpenViVQA [22] and the recent reasoning-oriented AutoViVQA [21] make it possible to measure progress in this low-resource setting.

Obtaining a strong model on a new Vietnamese benchmark, however, is still expensive. The common recipe fine-tunes an existing VLM: the fine-tuned Vintern-1B baseline of AutoViVQA adds LoRA adapters to the language model and splits every image into up to six high-resolution tiles plus a thumbnail, so that the decoder reads up to 1,792 visual tokens per image. The alternative is to design a task-specific architecture, as ViMoE-VQA [20] does with a sparse mixture-of-experts fusion module and a decoder trained on AutoViVQA. A cheaper option is to keep the whole VLM frozen and train only a small connector between its encoder and decoder. Existing connectors follow two designs, each with its own weakness. A *global-only bridge* maps the global image vector to a few tokens; it is cheap, but it loses local details such as small objects, printed text and spatial relations. A *learned patch compressor*, such as a Q-Former, keeps these details, but it has to learn a new mapping from visual features to the decoder, which is difficult with limited Vietnamese data.

Vintern-1B suggests a third option. Its projector was pre-trained to map image regions into the input space of the decoder, so this alignment already exists and does not need to be learned again. At the same time, the model discards the global `[CLS]` vector of its encoder, which summarises the whole scene. This raises our research question: *when labelled data is limited, is reusing the pre-trained visual alignment of a VLM, with pooling to control the number of tokens, more effective than learning a new connector, and how many global and local tokens does the frozen decoder need?*

To address this question, we propose **ViBridge-VQA**, which gives the frozen decoder of Vintern-1B two complementary kinds of visual tokens from a single image view. *Local* tokens reuse the frozen projector of Vintern-1B, average-pooled to a chosen budget and placed in its original image slot, so the decoder receives region features in the representation it was trained to read. *Global* tokens are produced from the `[CLS]` vector by a small trainable bridge and add the scene-level information that Vintern-1B ignores. Only this bridge is trained. Because every additional token increases inference cost, we treat the number of global and local tokens as a budget and select it as the knee of the accuracy–cost Pareto front. As a result, ViBridge-VQA with 14 global and 36 local tokens reaches an F1 comparable to the published fine-tuned Vintern-1B while its decoder reads 36 times fewer visual tokens and none of its weights is updated.

In summary, our main contributions are as follows:

- We propose **ViBridge-VQA**, a global–local bridge that improves a frozen Vietnamese VLM by combining trainable global tokens with local tokens from its own pre-trained projector. With 50 visual tokens, it raises the F1 of Vintern-1B on AutoViVQA from 17.6 (zero-shot) to 54.6 when only the bridge is trained (1.35% of the parameters), and to 58.2 when a lightweight decoder LoRA is trained with it (1.57%).
- We study the **visual token budget** systematically over eight configurations and three random seeds, and select it on the validation split as the knee of the accuracy–cost Pareto front. The choice is the same for every metric and for cost measured in tokens or in runtime.
- We **analyse where the gains come from**: adding local tokens improves F1 far more than increasing the global tokens from 8 to 14, local tokens from the pre-trained projector work better than a learned compressor, they complement decoder adaptation, and they improve results on four other Vietnamese VQA datasets.

2 Related Work

Early VQA systems classified over a fixed set of answers, whereas current generative VLMs produce free-form answers with an autoregressive language model [12,1,19]. Vietnamese datasets include ViVQA [30], OpenViVQA [22], ViCLEVR [32], ViOCRVQA [26] and ViTextVQA [24]. AutoViVQA [21] contributes 37,077 questions on 19,411 COCO images [17], generated by large language models, filtered by an ensemble of validators and annotated with reasoning categories, with an emphasis on relational, causal and multi-step reasoning. On the modelling side, BARTPhoBEiT [31] and AViVQA-TranConI [23] combine pre-trained visual and Vietnamese language representations, Vintern-1B [7] brings an instruction-tuned VLM to Vietnamese, and ViMoE-VQA [20] routes multimodal tokens to sparse experts for generative VQA on AutoViVQA. AutoViVQA and ViMoE-VQA are earlier work of our group; here we use the AutoViVQA data but no component of ViMoE-VQA, and the method, the image-disjoint split and all our results are new. These approaches either train a new model or fine-tune an existing VLM on multi-tile inputs; in contrast, we keep every pre-trained weight of Vintern-1B frozen and use a single image view.

Keeping the vision encoder and the language model frozen and training only a connector is a common way to build VLMs efficiently, and connectors differ mainly in what they must learn. Frozen [33] learns a visual prefix from a global image representation. The Q-Former of BLIP-2 [12] learns a set of queries that compress the patch features, which preserves details but requires a large module (188M parameters) pre-trained on millions of image–text pairs. LLaVA [19,18] projects every patch token with a small MLP, which preserves locality but passes hundreds of tokens to the decoder; Honeybee [4] proposes locality-preserving abstractors, and InternVL [6], on which Vintern-1B is built, merges neighbouring patches before its MLP projector. In all of these models, the mapping from patches to the decoder is learned during large-scale pre-training. ViBridge-VQA

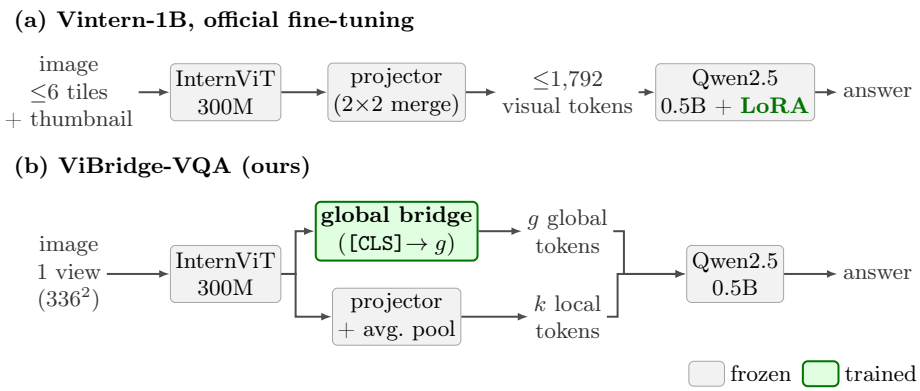


Fig. 1: Overview of ViBridge-VQA compared with the official fine-tuning of Vintern-1B. (a) The official recipe encodes up to seven views and adapts the decoder with LoRA. (b) ViBridge-VQA encodes one view; a trainable bridge maps the `[CLS]` vector to g global tokens, and the frozen projector gives k pooled local tokens (selected: $g = 14$, $k = 36$).

differs in two respects: it does not learn this local mapping again but reuses the projector already pre-trained in the VLM, and it trains only a small global bridge (12.9M parameters) on about 26k task questions.

The number of visual tokens largely determines the cost of the decoder. Inference-optimal scaling of VLMs [13] shows that a few tokens can suffice for many tasks, and Matryoshka multimodal models [3] train a model to work at several nested pooled token budgets and show that the required budget depends on the task. Other work reduces the visual tokens directly: DeCo [35] replaces the Q-Former with adaptive average pooling of the patch tokens and reports better results, TokenPacker [14] learns a compact projector, and LLaVA-PruMerge [29] and FastV [5] prune or merge visual tokens of an existing model. Like DeCo, we rely on simple average pooling, but we apply it to the output of a projector that is already pre-trained, keep the whole VLM frozen, and choose among budgets with explicit criteria, namely the knee of the Pareto front [28] and TOPSIS [11]. Parameter-efficient methods such as adapters [9], prefix tuning [15] and LoRA [10] reduce the cost of adaptation from the other side, by training a small number of additional weights in the decoder. We use decoder LoRA both as a baseline and as a complement, and ask whether a small training budget is better spent on the visual tokens or on adapting the decoder.

3 Proposed Method

We formulate VQA as conditional answer generation: given an image I and a Vietnamese question q, the model generates a short free-form answer $y = (y_1, \ldots, y_T)$. ViBridge-VQA builds on Vintern-1B [7] and keeps all of its pre-trained weights frozen. Instead of adapting the model, it changes what the decoder sees.

3.1 Overall Architecture

As illustrated in Fig. 1, ViBridge-VQA keeps the three components of Vintern-1B: the InternViT-300M vision encoder (24 layers, width 1024), its two-layer MLP projector P, and the Qwen2.5-0.5B-Instruct language decoder (24 layers, width 896). Each image is resized to one 336×336 view (24×24 patches of 14×14 pixels); InternViT, pre-trained at 448×448, bicubically interpolates its position embeddings to this resolution. The encoder returns one vector per patch and a global `[CLS]` vector $v \in \mathbb{R}^{1024}$. Vintern-1B uses only the patch vectors: it merges every 2×2 block of neighbouring patches into one 4096-dimensional vector (pixel shuffle) and maps each merged vector into the decoder input space with P, which was pre-trained to align the two backbones. At our resolution this yields a 12×12 grid of 144 region tokens. ViBridge-VQA uses both outputs of the encoder.

3.2 Global Tokens

The `[CLS]` vector summarises the whole scene, but Vintern-1B never passes it to the decoder. A small trainable bridge g_ϕ expands it into g tokens with two linear heads:

$$g_\phi(v) = \left[\, W_a v + b_a;\ \mathrm{reshape}_{(g-1)\times 896}(W_e v + b_e) \,\right] \in \mathbb{R}^{g \times 896}. \tag{1}$$

where $W_a \in \mathbb{R}^{896\times 1024}$ and $W_e \in \mathbb{R}^{(g-1)896\times 1024}$. The first head produces an *anchor* token, and the second produces $g-1$ complementary tokens that present the global content to the decoder from different learned directions. The bridge has $1025 \cdot 896 \cdot g$ parameters, i.e., 12,857,600 for $g = 14$, and its g tokens are placed before the prompt. We consider $g \in \{8, 14\}$: $g = 8$ is our global-only model, and $g = 14$ gave the highest F1 when the global-only bridge was varied from 4 to 20 tokens (Section 4.5). The final pair (g, k) is selected with the criterion of Section 3.5.

3.3 Local Tokens

To give the decoder region-level detail, ViBridge-VQA also passes local tokens, taken directly from the 144 region tokens of the frozen projector P. To control their number, we average-pool the 12×12 grid to an $s \times s$ grid, so that $k = s^2$ with $s \in \{1, 3, 6, 12\}$: $k = 144$ keeps every region token, and $k = 36$ averages each 2×2 block. The pooled tokens replace the embeddings of k image-context placeholders inside Vintern-1B's own image slot, `<img>` $\cdots$ `</img>`, in the prompt. Because they come from the pre-trained projector rather than from a newly learned projection of the patches, the local tokens stay in the representation that the decoder was trained to read, and no parameters are needed to produce them.

3.4 Training Objective

Only the bridge parameters ϕ are trained; the vision encoder, the projector and the decoder remain frozen, and no adapter is added to the decoder. We minimise

the cross-entropy of the answer tokens:

$$\mathcal{L} = -\sum_{t=1}^{T} \log p\big(y_t \mid y_{<t},\, g_\phi(v),\, \mathrm{pool}_s(P(x)),\, q\big), \tag{2}$$

where x denotes the merged patch vectors. ViBridge-VQA can also be combined with decoder adaptation: in that variant, a rank-16 LoRA [10] on the query, key, value and output projections of every decoder layer is trained jointly with the bridge, which adds 2,162,688 parameters. Throughout the paper, percentages of trained parameters are relative to all parameters of the model, i.e., the frozen components of Vintern-1B (about 0.94B, including the vision encoder, projector, embeddings and output layer) plus the trained modules. With $g = 14$, the bridge alone trains 12.86M parameters (1.35%), and the bridge with LoRA 15.02M (1.57%).

3.5 Choosing the Token Budget

The cost of the decoder grows with the number of visual tokens $g + k$, so we treat the pair (g, k) as a budget to be chosen. In our experiments, accuracy increases monotonically with the number of visual tokens, so every configuration lies on the accuracy–cost Pareto front and the front alone does not single one out. We therefore select its *knee* [28], the point after which additional tokens bring diminishing returns. After normalising cost c and accuracy a to $[0, 1]$ over the front, the knee is the configuration that maximises $\hat{a} - \hat{c}$, i.e., the one lying furthest above the line joining the cheapest and the most accurate configuration. The knee is computed on the validation split only, and we check the choice with the multi-criteria method TOPSIS [11], also on validation. Test results are computed for every configuration only to be reported; no choice is based on them.

4 Experiments

4.1 Dataset and Evaluation Metrics

Dataset. We evaluate on AutoViVQA [21], a Vietnamese VQA benchmark on MS COCO images [17] in which each question has five short free-form reference answers and a reasoning type. To measure answers about images the model has never seen, we use an image-disjoint split. After removing 273 duplicated (image, question) pairs and 97 questions without one of the eight reasoning labels, we partition the 36,707 questions by image into a 70/15/15 split (seed 42), so that no image appears in more than one split: 25,776 / 5,463 / 5,468 questions on 13,579 / 2,909 / 2,910 images. In a preliminary comparison, this split lowered the F1 of the global-only bridge by about 0.5 points relative to a split that does not group questions by image.

Evaluation metrics. Following AutoViVQA and ViMoE-VQA, we report exact-match Accuracy, word-level Precision, Recall and F1, BLEU [25], ROUGE-L [16], METEOR [2] and CIDEr [34], all against the five reference answers and multiplied by 100. All our models are scored with the same implementation.

4.2 Experimental Setup

Implementation details. The vision encoder, the projector and the decoder are initialised from Vintern-1B-v3.5 [7] and kept frozen. ViBridge-VQA is trained for two epochs, after which neither the validation loss nor CIDEr improves further; the variants with a decoder LoRA are trained for one epoch, and the global-only model with LoRA also for three epochs, its strongest setting (+1.3 validation F1). We use AdamW with a learning rate of 2×10^{-4} and an effective batch size of 8, train on the first reference answer, and decode greedily (batch size 2, at most 50 new tokens). Unless stated otherwise, every configuration is trained with three random seeds (42, 123 and 3407), and we report the mean and the standard deviation. Every model fits on a single 16 GB GPU such as an NVIDIA T4. The token budget is selected on the validation split, and the test split is used only for the final report.

Baselines. We compare against three groups of systems: (1) multimodal models fine-tuned on AutoViVQA, namely ViT5_ViT, BARTPhoBEiT [31], Vintern-1B and ViMoE-VQA [20]; (2) models used zero-shot, namely Vintern-1B, LLaMA 3.2, Gemini 2.0 and 2.5 Flash, and GPT-5; and (3) our *global-only* model, which uses the same bridge without local tokens ($k = 0$, $g = 8$), with and without a decoder LoRA. The results of groups (1) and (2) are those reported in the AutoViVQA and ViMoE-VQA papers. The fine-tuned Vintern-1B follows the official recipe: the vision encoder and the projector are frozen, a rank-16 LoRA adapts the language model for one epoch, and each image is encoded as up to six 448×448 tiles plus a thumbnail.

4.3 Main Results

Table 1 compares ViBridge-VQA with the baselines. Without updating any weight of Vintern-1B, ViBridge-VQA raises its F1 from 17.55 in the zero-shot setting to 54.56. With the selected budget of 50 visual tokens, its precision, recall and F1 are comparable to the published values of the fine-tuned Vintern-1B, while its exact-match accuracy is 1.3 points lower; with all 144 local tokens, its accuracy is also comparable (13.19 versus 13.01). Its decoder reads 50 or 158 visual tokens, 36 and 11 times fewer than the 1,792 of the fine-tuned Vintern-1B.

Decoder adaptation adds to these gains rather than replacing them. Training ViBridge-VQA jointly with a rank-16 LoRA for a single epoch raises test F1 by a further 3.6 points to 58.17 and gives the highest exact-match accuracy in Table 1. The task-specific ViMoE-VQA, whose fusion module and decoder are trained on AutoViVQA, keeps the highest precision and F1, 2.5 F1 points above our best model. ViBridge-VQA reaches its results differently: instead of designing and

Table 1: Performance on the AutoViVQA test set (×100). Baselines: results reported in AutoViVQA [21] and ViMoE-VQA [20] (ViMoE-VQA: mean of five seeds). Ours: ViBridge-VQA ($g = 14$, $k = 36$, 50 visual tokens) without and with a one-epoch decoder LoRA, image-disjoint split (Section 4.1), mean over three seeds, share of trained parameters in parentheses; standard deviations in Section 4.3. R-L: ROUGE-L, MET.: METEOR. Bold: best per column.

	Model	Acc	P	R	F1	BLEU	R-L	MET.	CIDEr
Fine-tuned	ViT5_ViT	7.97	46.84	50.33	48.52	4.13	46.89	31.02	72.68
	BARTPhoBEiT	8.81	45.30	46.48	45.88	4.33	44.83	24.57	**188.96**
	Vintern-1B	13.01	52.47	55.12	53.76	6.11	51.93	35.25	72.84
	ViMoE-VQA	9.55	**62.89**	58.60	**60.69**	12.54	47.12	39.10	88.76
Zero-shot	Vintern-1B	0.12	17.52	19.87	17.55	1.91	25.84	23.93	8.54
	LLaMA 3.2	0.36	23.96	73.71	36.16	3.62	36.11	30.01	62.84
	Gemini 2.0 F.	0.55	27.20	74.10	39.79	4.41	39.60	31.72	74.42
	Gemini 2.5 F.	0.22	24.43	**76.66**	24.75	0.39	37.27	31.22	71.90
	GPT-5	10.84	47.20	55.20	50.89	6.07	47.30	33.34	84.20
Ours	ViBridge-VQA (1.35%)	11.75	55.36	56.56	54.56	18.52	52.64	45.53	110.58
	ViBridge-VQA + LoRA (1.57%)	**14.67**	59.02	59.96	58.17	**22.53**	**56.25**	**49.32**	122.88

training a new architecture, it reuses an existing VLM and trains 1.57% of its parameters, and it leads in exact-match accuracy.

Compared with our global-only model (Table 3, Fig. 2a), ViBridge-VQA with a frozen decoder already reaches the global-only model trained with a three-epoch decoder LoRA (54.96 versus 54.78 validation F1), and with the same one-epoch LoRA it is 4.7 F1 above it (58.21 versus 53.47).

Statistical stability. On the test split, the standard deviation of ViBridge-VQA is at most 0.19 for F1 and 0.95 for CIDEr, and for both models in Table 1, test F1 lies at most 0.6 points below validation F1.

4.4 Choosing the Token Budget

Table 2 and Fig. 2a vary the number of global tokens $g \in \{8, 14\}$ and of local tokens $k \in \{1, 9, 36, 144\}$. The most striking pattern is that *adding local tokens improves F1 far more than adding global tokens.* At $g = 14$, increasing k from 1 to 144 adds 4.9 F1 and 19.1 CIDEr on the validation split, whereas increasing g from 8 to 14 at a fixed k adds only 0.2–0.6 F1. Accuracy has not saturated within the 144 region tokens available at our resolution, as each step of k still adds 1.3–2.0 F1, but the gain per token drops quickly, from 0.07 F1 per token between 23 and 50 tokens to 0.015 between 50 and 158. The budget is therefore a genuine trade-off between accuracy and cost.

Because accuracy grows with every added token, all configurations lie on the Pareto front, and we select its knee (Section 3.5). On the validation split, the knee is $g = 14$, $k = 36$, and it is the same for each of the eight metrics, whether cost is counted in visual tokens or as evaluation runtime. As a post-hoc check that played no role in the selection, the knee computed on the test split falls on the same configuration. TOPSIS on validation, with equal weight on accuracy

Table 2: Effect of the token budget (×100, mean over three seeds, ± standard deviation for F1; global only: four seeds). Tokens: visual tokens passed to the decoder ($g + k$). Time: evaluation runtime (minutes) of one session that loads the model and answers the 10,931 validation and test questions (greedy, batch 2), mean over seeds (Kaggle GPUs). The selected budget is in bold.

				Validation		Test	
g	k	Tokens	Time	F1	CIDEr	F1	CIDEr
8	0	8	–	50.74 ±.17	99.07	50.47 ±.22	96.19
14	0	14	–	50.67 ±.08	99.83	–	–
8	1	9	57.0	51.03 ±.12	99.27	50.71 ±.16	96.90
8	9	17	59.2	52.82 ±.11	106.11	52.11 ±.25	101.30
8	36	44	60.6	54.32 ±.41	111.88	53.82 ±.22	107.67
8	144	152	71.2	56.15 ±.08	119.05	55.20 ±.07	114.13
14	1	15	58.8	51.67 ±.25	101.66	51.67 ±.12	99.53
14	9	23	57.7	52.98 ±.08	106.02	52.76 ±.28	103.63
14	**36**	**50**	**60.0**	**54.96** ±.28	**113.29**	**54.56** ±.19	**110.58**
14	144	158	70.2	56.61 ±.04	120.71	56.04 ±.19	116.74

and cost, makes the same choice. It obtains 72% of the F1 gain of $k = 144$ over global only with 32% of its tokens. It is a compromise between accuracy and cost, not the most accurate configuration: $k = 144$ is 1.7 F1 higher. $g = 14$ is better than $g = 8$ at every k.

Computational cost. We distinguish three kinds of cost. *Decoder input*: with the official tiling (at most six tiles plus a thumbnail), every AutoViVQA test image is split into six tiles plus a thumbnail, i.e., 1,792 visual tokens, against 50 for ViBridge-VQA. *Vision encoding*: InternViT-300M processes 577 tokens for one 336×336 view, about 191 GMACs, against 1,025 tokens and 362 GMACs for each 448×448 view (analytic counts, matching fvcore at 448×448); with seven views, Vintern-1B thus needs 2,533 GMACs per image, 13 times more. *Evaluation runtime*: within our own budgets, keeping all 144 local tokens instead of 36 makes an evaluation session about 17% longer (Table 2).

4.5 Ablation Study

Where should the visual tokens come from? Table 3 compares three sources of visual tokens: the global `[CLS]` vector (a one-token *Residual* bridge or the *Multi-Token* bridge of Eq. (1)), a learned compression of the patches (*Tile-Attention* with eight learned queries, or a light or full Q-Former [12]), and our local tokens. Without decoder adaptation, the Multi-Token bridge is the best of these five, and the bridges that compress the patches are 3.7–5.6 F1 lower despite having up to 9.4× more parameters. Taking local tokens from the pre-trained projector instead works better: at the same $g = 8$, adding 36 or 144 of them improves F1 by 3.6 and 5.4 points (Table 2).

Local tokens and decoder adaptation are complementary. A decoder LoRA improves every global-only bridge and brings them within 0.8 F1 of one another

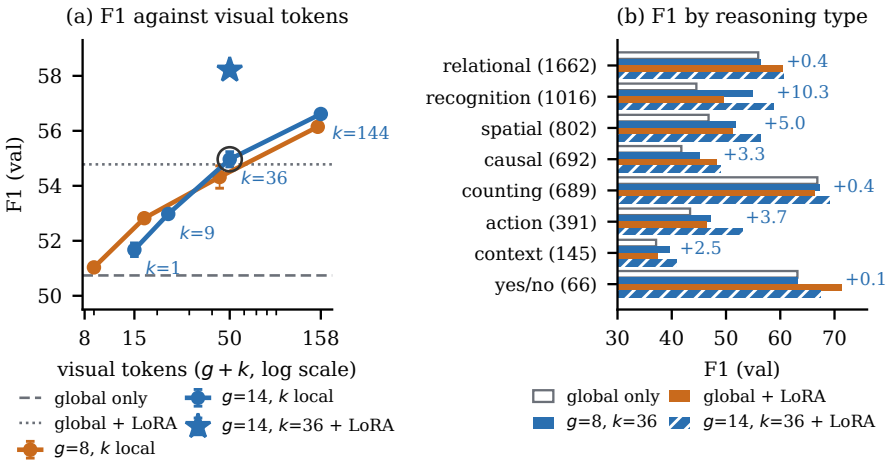


Fig. 2: (a) Validation F1 against the number of visual tokens (mean ± std over three seeds); the circled point is the knee, $g = 14$, $k = 36$. Horizontal lines: the global-only model (four seeds) and the same model with a three-epoch decoder LoRA (three seeds). (b) Validation F1 by reasoning type (number of questions in parentheses), each model averaged over all of its seeds; the numbers give the gain of adding 36 local tokens at the same $g = 8$. Global + LoRA: three-epoch decoder LoRA; $g = 14$, $k = 36$ + LoRA: ViBridge-VQA with a one-epoch decoder LoRA.

(52.69–53.47). Even after this adaptation, all of them remain below ViBridge-VQA with a frozen decoder. Decoder adaptation is therefore not needed to surpass them, but it is still useful: in the same one-epoch joint training, it adds 3.3 F1 to ViBridge-VQA, more than the 2.7 F1 it adds to the Multi-Token bridge. Distilling the projector into the global-only bridge instead lowers F1 by 0.3–0.4 points.

Number of global tokens. We varied the Multi-Token bridge from $g = 4$ to $g = 20$ tokens (three seeds each). Validation F1 rose from 48.5 at $g = 4$ to 49.6 at $g = 8$ and 50.7 at $g = 14$, and stayed between 50.4 and 50.6 for $g = 16$–20.

4.6 Analysis by Reasoning Type and Error Analysis

Reasoning types. Fig. 2b breaks the results down by reasoning category. At the same $g = 8$, adding 36 local tokens helps most where the answer depends on what is visible in the image: recognition (+10.3 F1, Fig. 2b), spatial (+5.0) and action (+3.7) questions. On these categories, this frozen-decoder model is also better than the global-only model with a decoder LoRA. Conversely, decoder adaptation is more useful for causal and relational questions, whose reference answers are the longest (5.0 and 5.6 words on average), and for yes/no questions. Adding the LoRA to ViBridge-VQA combines both strengths: the resulting model is the best of the four models in Fig. 2b in seven of the eight categories, slightly above the global-only model with LoRA on causal (48.9 versus 48.3 F1) and relational

Table 3: Where the visual tokens come from, with and without decoder adaptation (validation, ×100, mean over three seeds with ± std for F1; Multi-Token: four seeds from its original runs; *run with seed 42 only). Bridge only: two epochs; + LoRA: rank-16 decoder LoRA (2.16M) trained jointly for one epoch. Params: bridge parameters.

			Bridge only		+ decoder LoRA	
Visual tokens from	Tokens	Params	F1	CIDEr	F1	CIDEr
Global vector (`[CLS]`)						
Residual	1	4.86M	46.02 ±.36	86.48	52.69 ±.14	104.25
Multi-Token	8	7.35M	50.74 ±.17	99.07	53.47 ±.17	106.68
Learned compression of the patch tokens						
Tile-Attention	8	4.14M	45.14 ±.92	84.14	52.82*	104.72*
Light Q-Former	8	27.6M	47.06 ±.66	88.07	53.21 ±.14	106.26
Full Q-Former	16	69.4M	46.84 ±.47	87.74	53.29 ±.29	106.50
Global vector + pre-trained projector (ViBridge-VQA, $g = 14$)						
$k = 36$	50	12.86M	54.96 ±.28	113.29	**58.21** ±.16	**124.88**
$k = 144$	158	12.86M	**56.61** ±.04	**120.71**	–	–

Table 4: Evaluation on Vietnamese VQA datasets not used to train our models (×100; three random samples of about 1,000 test questions, 505–519 for ViVQA; mean, ± std for F1 over the samples, with one trained model per configuration, seed 42). Vintern-1B reads up to six tiles plus a thumbnail; the other models are trained on AutoViVQA only (seed 42). Best results per column are in bold.

	ViTextVQA		OpenViVQA		ViVQA-X		ViVQA	
Model	F1	CIDEr	F1	CIDEr	F1	CIDEr	F1	CIDEr
Vintern-1B, zero-shot	32.01 ±.31	**195.62**	**58.40** ±.80	**411.63**	14.52 ±.47	50.13	23.90 ±.24	78.63
Global only + LoRA, 3 ep.	3.96 ±.09	15.64	21.56 ±.48	83.21	15.89 ±.65	42.12	31.48 ±1.15	84.94
ViBridge-VQA, k=36	15.14 ±.47	66.39	28.24 ±.79	128.36	24.91 ±.14	79.51	42.68 ±1.52	171.21
ViBridge-VQA, k=144	**33.26** ±.67	190.30	35.89 ±.80	187.00	**27.89** ±.13	92.22	**44.95** ±.90	**191.28**
ViBridge-VQA, k=36 + LoRA	15.39 ±.39	70.07	28.99 ±.51	130.46	27.64 ±.11	**95.29**	44.76 ±1.21	183.60

(60.5 versus 60.4) questions and well above it on recognition (58.7 versus 49.6); the only exception is yes/no questions.

Error analysis. On the validation predictions (all seeds), our answers are as long as the references (4.3 words on average). Most predictions (77–81%) share some, but not all, words with the closest reference. Compared with the global-only model, ViBridge-VQA increases the share of predictions with F1 = 1 from 9.7% to 13.3% and reduces the share with no word in common with any reference from 9.6% to 8.0% (6.8% with LoRA). Complete failures are most frequent on context, recognition, spatial and action questions (12–18%).

4.7 Generalisation to Other Datasets

Table 4 evaluates the models without further training on four Vietnamese VQA datasets: two scene-text datasets, ViTextVQA [24] and OpenViVQA [22], and two general-domain ones, ViVQA-X [8] and ViVQA [30]. The strongest global-only model (with a three-epoch LoRA) serves as a reference. Local tokens change the



Fig. 3: Qualitative examples from the AutoViVQA validation set (seed 42). Q: question, GT: one of the five reference answers; Ours: ViBridge-VQA ($g = 14$, $k = 36$), Ours+LoRA: ViBridge-VQA with a one-epoch decoder LoRA.

picture most on scene text, where the global-only model collapses. On ViTextVQA, F1 rises from 3.96 to 15.14 with 36 local tokens and to 33.26 with 144, on par with the zero-shot Vintern-1B (32.01) that reads six 448×448 tiles, although ViBridge-VQA reads a single 336×336 view. On OpenViVQA, whose references are long descriptive sentences, local tokens narrow the gap to the six-tile Vintern-1B.

On the general-domain datasets, ViBridge-VQA beats both references, by 10.4 (ViVQA-X) and 18.8 (ViVQA) F1 over Vintern-1B with $k = 36$. Adding the decoder LoRA to ViBridge-VQA helps on these general-domain datasets, bringing $k = 36$ close to $k = 144$, but barely on scene text. On ViTextVQA, 144 local tokens give 33.26 F1, whereas 36 local tokens with LoRA give 15.39.

4.8 Qualitative Analysis

Fig. 3 illustrates how the two components work together. In every example, ViBridge-VQA already identifies the relevant content of the image (the driver's phone, the helmets, the football match, the bird feeder), but its answer remains vague. Adding the decoder LoRA turns it into the precise answer, for instance that the driver is using *the phone* and that the helmets are worn *to protect their heads*, in line with the gains of decoder adaptation on causal and yes/no questions (Section 4.6).

5 Conclusion

In this work, we introduced **ViBridge-VQA**, a parameter-efficient approach that improves the Vietnamese VLM Vintern-1B for generative VQA without updating any of its pre-trained weights, by giving its frozen decoder trainable global tokens from the `[CLS]` vector and local tokens from the pre-trained projector. With 14 global and 36 local tokens, the knee of the accuracy–cost Pareto front on validation, training only the bridge raises F1 on AutoViVQA from 17.6 to 54.6, comparable to the published F1 of the fine-tuned Vintern-1B, and training a lightweight decoder LoRA with it raises F1 to 58.2. Adding local tokens improves F1 far more than increasing the global tokens from 8 to 14, and they complement decoder adaptation.

The task-specific ViMoE-VQA still reaches a higher F1, and accuracy had not saturated at the 144 local tokens available at our resolution.

Overall, an existing Vietnamese VLM can be improved substantially by choosing which visual tokens its frozen decoder receives, and how many, with an explicit accuracy–cost criterion. The code and the image-disjoint split are available at `https://github.com/nguyennn263/modular-vlm-finetune`.

Acknowledgments. This research is funded by University of Science, VNU-HCM under grant number CNTT 2025-02.

References

1. Alayrac, J.B., Donahue, J., Luc, P., Miech, A., Barr, I., Hasson, Y., Lenc, K., Mensch, A., Millican, K., Reynolds, M., et al.: Flamingo: a visual language model for few-shot learning. In: Advances in Neural Information Processing Systems (NeurIPS). pp. 23716–23736 (2022)
2. Banerjee, S., Lavie, A.: METEOR: An automatic metric for MT evaluation with improved correlation with human judgments. In: Proceedings of the ACL Workshop on Intrinsic and Extrinsic Evaluation Measures for Machine Translation and/or Summarization. pp. 65–72 (2005)
3. Cai, M., Yang, J., Gao, J., Lee, Y.J.: Matryoshka multimodal models (2024), arXiv preprint arXiv:2405.17430
4. Cha, J., Kang, W., Mun, J., Roh, B.: Honeybee: Locality-enhanced projector for multimodal LLM. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 13817–13827 (2024)
5. Chen, L., Zhao, H., Liu, T., Bai, S., Lin, J., Zhou, C., Chang, B.: An image is worth 1/2 tokens after layer 2: Plug-and-play inference acceleration for large vision-language models (2024), arXiv preprint arXiv:2403.06764
6. Chen, Z., Wu, J., Wang, W., Su, W., Chen, G., Xing, S., Zhong, M., Zhang, Q., Zhu, X., Lu, L., Li, B., Luo, P., Lu, T., Qiao, Y., Dai, J.: InternVL: Scaling up vision foundation models and aligning for generic visual-linguistic tasks. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 24185–24198 (2024)
7. Doan, K.T., Huynh, B.G., Hoang, D.T., Pham, T.D., Pham, N.H., Nguyen, Q.T.M., Vo, B.Q., Hoang, S.N.: Vintern-1B: An efficient multimodal large language model for Vietnamese (2024), arXiv preprint arXiv:2408.12480

8. Duong, T.B., Tran, H.M., Le-Nguyen, B.N., Duong, D.T.: An automated pipeline for constructing a Vietnamese VQA-NLE dataset. In: Proceedings of the International Conference on Intelligent Systems and Networks (ICISN). pp. 164–173. Lecture Notes in Networks and Systems, Springer (2026)
9. Houlsby, N., Giurgiu, A., Jastrzebski, S., Morrone, B., de Laroussilhe, Q., Gesmundo, A., Attariyan, M., Gelly, S.: Parameter-efficient transfer learning for NLP. In: Proceedings of the International Conference on Machine Learning (ICML). pp. 2790–2799 (2019)
10. Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W.: LoRA: Low-rank adaptation of large language models. In: Proceedings of the International Conference on Learning Representations (ICLR) (2022)
11. Hwang, C.L., Yoon, K.: Multiple Attribute Decision Making: Methods and Applications. Springer (1981)
12. Li, J., Li, D., Savarese, S., Hoi, S.: BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In: Proceedings of the International Conference on Machine Learning (ICML). pp. 19730–19742 (2023)
13. Li, K.Y., Goyal, S., Semedo, J.D., Kolter, J.Z.: Inference optimal VLMs need only one visual token but larger models (2024), arXiv preprint arXiv:2411.03312
14. Li, W., Yuan, Y., Liu, J., Tang, D., Wang, S., Qin, J., Zhu, J., Zhang, L.: TokenPacker: Efficient visual projector for multimodal LLM (2024), arXiv preprint arXiv:2407.02392
15. Li, X.L., Liang, P.: Prefix-tuning: Optimizing continuous prompts for generation. In: Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL). pp. 4582–4597 (2021)
16. Lin, C.Y.: ROUGE: A package for automatic evaluation of summaries. In: Proceedings of the ACL Workshop on Text Summarization Branches Out. pp. 74–81 (2004)
17. Lin, T.Y., Maire, M., Belongie, S., Bourdev, L., Girshick, R., Hays, J., Perona, P., Ramanan, D., Zitnick, C.L., Dollár, P.: Microsoft COCO: Common objects in context. In: Proceedings of the European Conference on Computer Vision (ECCV). pp. 740–755 (2014)
18. Liu, H., Li, C., Li, Y., Lee, Y.J.: Improved baselines with visual instruction tuning. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 26296–26306 (2024)
19. Liu, H., Li, C., Wu, Q., Lee, Y.J.: Visual instruction tuning. In: Advances in Neural Information Processing Systems (NeurIPS). pp. 34892–34916 (2023)
20. Nguyen, A.T., Nguyen, T.Q., Dang, D.L., Tran, D.T., Phan, B.D., Nguyen, Q.T., Le, T.: ViMoE-VQA: Reasoning-aware sparse experts for generative Vietnamese visual question answering. Procedia Computer Science (2026), to appear
21. Nguyen, A.T., Phan, B.D., Nguyen, T.Q., Tran, D.T., Dang, D.L., Nguyen, Q.T., Le, T.: AutoViVQA: A large-scale, automatically constructed dataset for Vietnamese visual question answering. In: Proceedings of the Asian Conference on Intelligent Information and Database Systems (ACIIDS). pp. 492–506. Communications in Computer and Information Science, Springer (2026)
22. Nguyen, N.H., Vo, D.T.D., Nguyen, K.V., Nguyen, N.L.T.: OpenViVQA: Task, dataset, and multimodal fusion models for visual question answering in Vietnamese. Information Fusion **100**, 101868 (2023)
23. Nguyen, N.S., Nguyen, V.S., Le, T.: Advancing Vietnamese visual question answering with transformer and convolutional integration. Computers and Electrical Engineering **119**, 109474 (2024)

24. Nguyen, Q.V., Tran, D.Q., Pham, H.Q., Nguyen, T.K.B., Nguyen, N.H., Nguyen, K.V., Nguyen, N.L.T.: ViTextVQA: A large-scale visual question answering dataset for evaluating Vietnamese text comprehension in images (2025), arXiv preprint arXiv:2404.10652
25. Papineni, K., Roukos, S., Ward, T., Zhu, W.J.: BLEU: a method for automatic evaluation of machine translation. In: Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL). pp. 311–318 (2002)
26. Pham, H.Q., Nguyen, T.K.B., Nguyen, Q.V., Tran, D.Q., Nguyen, N.H., Nguyen, K.V., Nguyen, N.L.T.: ViOCRVQA: Novel benchmark dataset and vision reader for visual question answering by understanding Vietnamese text in images (2024), arXiv preprint arXiv:2404.18397
27. Qwen Team: Qwen2.5 technical report (2024), arXiv preprint arXiv:2412.15115
28. Satopää, V., Albrecht, J., Irwin, D., Raghavan, B.: Finding a "kneedle" in a haystack: Detecting knee points in system behavior. In: Proceedings of the International Conference on Distributed Computing Systems Workshops (ICDCSW). pp. 166–171 (2011)
29. Shang, Y., Cai, M., Xu, B., Lee, Y.J., Yan, Y.: LLaVA-PruMerge: Adaptive token reduction for efficient large multimodal models (2024), arXiv preprint arXiv:2403.15388
30. Tran, K.Q., Nguyen, A.T., Le, A.T.H., Nguyen, K.V.: ViVQA: Vietnamese visual question answering. In: Proceedings of the Pacific Asia Conference on Language, Information and Computation (PACLIC). pp. 683–691 (2021)
31. Tran, K.V., Nguyen, K.V., Nguyen, N.L.T.: BARTPhoBEiT: Pre-trained sequence-to-sequence and image transformers models for Vietnamese visual question answering. In: Proceedings of the International Conference on Multimedia Analysis and Pattern Recognition (MAPR). pp. 1–6 (2023)
32. Tran, K.V., Phan, H.P., Nguyen, K.V., Nguyen, N.L.T.: ViCLEVR: A visual reasoning dataset and hybrid multimodal fusion model for visual question answering in Vietnamese (2023), arXiv preprint arXiv:2310.18046
33. Tsimpoukelli, M., Menick, J., Cabi, S., Eslami, S.M.A., Vinyals, O., Hill, F.: Multimodal few-shot learning with frozen language models. In: Advances in Neural Information Processing Systems (NeurIPS). pp. 200–212 (2021)
34. Vedantam, R., Zitnick, C.L., Parikh, D.: CIDEr: Consensus-based image description evaluation. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 4566–4575 (2015)
35. Yao, L., Li, L., Ren, S., Wang, L., Liu, Y., Sun, X., Hou, L.: DeCo: Decoupling token compression from semantic abstraction in multimodal large language models (2024), arXiv preprint arXiv:2405.20985